

HW08

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Exercise 1

Collaborators: Sam

Tools and resources used

Exercise 2

- Points attempted: 1
- Self-assessed score: 1/1.
- Time spent: 2 minutes.
- Difficulty: 1/10.
- Questions/Feedback: None..

HW8 Exercise 2: One thing that I would like to see on HW 9 would be some more work on burnside's theorem. Although I "know" the theorem, I do not have a full intuition for it yet.

Choice Exercise 5

- Points attempted: 10
- Self-assessed score: 10/10.
- Time spent: 30 minutes.
- Difficulty: 5/10.
- Questions/Feedback: None..

We defined a (left) group action of the group $(G, *)$ on the set X as a function $\cdot : G \times X \rightarrow X$ such that for all $x \in X$ and $g_1, g_2 \in G$

- $e \cdot x = x$, and
- $g_1 \cdot (g_2 \cdot x) = (g_1 * g_2) \cdot x$.

1. [4] For each of the following subparts, give an example of a group $(G, *)$, a set X , and a function $\cdot : G \times X \rightarrow X$ such that

(a) The function is a group action (i.e. it satisfies both of the above properties.)

We can have the group $SO(3)$ acting on the set of 3D vectors $\mathbb{R}^3$, where the function is a matrix-vector multiplication. The identity matrix $I \in SO(3)$ satisfies the first condition, and the associativity of matrix-vector products satisfy the second condition.

(b) The function satisfies the first property, but not the second.

We can have the group C_2 and the set $\mathbb{Z}$, with the function being addition, such that the identity in C_2 corresponds to 0 and r corresponds to 1. Now, the first condition is always met, however the second condition breaks.

(c) The function satisfies the second property, but not the first.

Now, we can have the group C_2 and the set $\mathbb{Z}/2\mathbb{Z}$, with the function

$$c \cdot x = \begin{cases} 1 + x & c = e \\ x & c = r \end{cases},$$

for $c \in C_2$ and $x \in \mathbb{Z}/2\mathbb{Z}$.

(d) We can The function satisfies neither property.

We can have any non-trivial group G on any set X such that $|X| > 1$, where the function always maps to some $x \in X$. This function breaks both conditions.

2. [3] Prove that if G is a group that acts on a set X , then the set

$$H = \{g \in G \mid g \cdot x = x \text{ for all } x \in X\}$$

is a normal subgroup of G .

Proof. To begin, we note that for H to be a normal subgroup of G , H itself must be a group. So, we must show that H satisfies the group axioms. First, associativity is inherited. Also notice that, by definition, $e \in H$.

Then, H is closed because for some $h, h' \in H$, $h \cdot x = x$ and $h' \cdot x = x$, so $h \cdot (h' \cdot x) = x$. This means that $(h * h') \cdot x = x$, so $h * h' \in H$. Thus, H is closed.

Now, we must prove that for all $h \in H$, $h^{-1} \in H$. First, we know that $e \in H$, because $e \cdot x = x$. Notice that $h^{-1} * h = e$, so $(h^{-1} * h) \cdot x = x$. By definition of a group action, $(h^{-1} * h) \cdot x = h^{-1} \cdot h \cdot x = h^{-1} \cdot x = x$. Thus, $h^{-1} \in H$.

Now, to show that $H \triangleleft G$, we must show that for any h, g , $g * h * g^{-1} \in H$. Let use directly apply this expresion onto some $x \in X$ to get $g \cdot h \cdot g^{-1} \cdot x$. Notice that $g^{-1} \cdot x \in X$, and $h \cdot x' = x'$ for any $x' \in X$. So, we can effectively remove h from the expression to get $g \cdot g^{-1} \cdot x = (g * g^{-1}) \cdot x = x$. Thus, $g * h * g^{-1} \in H$, so $H \triangleleft G$. $\square$

3. [3] Let $G = \mathbb{Z}$, $X = (H, *)$, and $h \in H$ be an element of H . Prove that the map $n \cdot h = h^n * h * h^{-n}$ is a group action.

Proof. First, we prove that $e \cdot x = x$. Let $e \in G = 0$. Now, $e \cdot h = h^0 * h * h^0 = h$. This satisfies the first condition.

To prove the second condition, let $n, n' \in G$, so

$$n \cdot (n' \cdot h) = n \cdot (h^{n'} * h * h^{-n'}) \quad (1)$$

$$= h^n * h^{n'} * h * h^{-n'} * h^{-n} \quad (2)$$

$$= h^{n+n'} * h * h^{-(n+n')} \quad (3)$$

$$= (n + n') \cdot h. \quad (4)$$

Therefore, the map $n \cdot h = h^n * h * h^{-n}$ is a group action. $\square$

Choice Exercise 7

1. [3] Set a timer for 25 minutes and read an algebra text of your choosing, attempting problems/exercises along the way.]

I chose to read chapter 6, in part 3 of the Infinitely Large Napkin, which covers the concept of the metric space. I was able to do 3 exercises in total.

Exercise 6.1.3. Use the triangle inequality to show these are equivalent. (The names “radius” and “diameter” are a big hint!)

A metric space $M = (M, d)$ is bounded if there is a constant D such that $d(p, q) \leq D$ for all $p, q \in M$.

$$d(p, r) \leq d(p, q) + d(q, r) \quad (5)$$

$$d(p, q) \leq R, d(q, r) \leq R \quad (6)$$

$$d(p, r) \leq 2R = D. \quad (7)$$

Exercise 6.1.6. Show that “totally bounded” implies “bounded”.

If a space is totally bounded, then it can be covered by finitely many ε -neighborhoods, for $\varepsilon > 0$.

Therefore, in the worst case, our diameter D will be $n\varepsilon$, where n is the number of ε -neighborhoods.

Question 6.2.2. Show that a sequence which converges is automatically Cauchy.

A sequence $x_1, x_1, \dots$ in a metric space M is Cauchy if for any $\varepsilon > 0$, $d_M(x_m, x_n) < \varepsilon$, for all sufficiently large m and n .

I will borrow an old definition of converges from Intro to Proofs:

A sequence $y_1, y_2, \dots \in \mathbb{R}$ converges to c if

$$\forall \varepsilon > 0, \exists m \in \mathbb{N}, \forall n > m, |y_n - c| < \varepsilon.$$

We have a couple moving peices here. In our defintion for convergence, we can view $\mathbb{R}$ as a metric space, where $d(p, q) = |p - q|$. Also, this definition only tells us the distance between our sequence at index m and some constant c , not the distance between two items y_n, y_m . However, by the triangle inequality, we can see that, given $d(y_n, c) < \varepsilon$ and $d(y_{n'}, c) < \varepsilon$ for $n' > n$, $d(y_n, y_{n'}) < d(y_n, c) + d(y_{n'}, c) = 2\varepsilon$. This satisfies our definition of Cauchy.